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Presentability §

the six functors

Last time Proved that for M an abelian group, the Fourier transform

$$\Phi_M: \text{Fun}(M, \text{Sp}) \longrightarrow \text{QCoh}(\text{BD}_M)$$

is an equivalence. Also showed that for a cone $\sigma \in N_{\mathbb{R}}$,

$$\Phi_\sigma: \text{Fun}(\Theta(\sigma)^\vee, \text{Sp}) \longrightarrow \text{QCoh}(X_\sigma/T)$$

is an equivalence.

Today Start constructing (compatible) lax sym. mon functors

$$\text{Fun}(\Theta(\sigma)^\vee, \text{Sp}) \longrightarrow \text{Sh}(M_{\mathbb{R}}, \text{Sp})$$

These will land in modules over an idempotent algebra w_σ , and be sym mon with this target.

> To do so, we need the six-functor formalism. So we'll start by reviewing that.

> It'll be convenient to understand a bit about presentable ∞ -categories.

Presentable ∞ -categories

Idea. Want a well-behaved class of large ∞ -categories with colimits that are controlled by a small amount of data.

Def. An infinite cardinal κ is regular if no set of cardinality κ is the union of fewer than κ sets of cardinality $< \kappa$.

Ex. The smallest infinite cardinal $\omega = \aleph_0$ is regular.

Def. Let κ be a regular cardinal. An ∞ -category I is κ -filtered if for every κ -small ∞ -category K , every diagram $K \rightarrow I$ admits an extension $K^\Delta \rightarrow I$.

If $\kappa = \omega$, we usually just say filtered instead of ω -filtered.

Note. When $\kappa = \omega$, this just says every finite diagram has a cone under it.

Exercise. Check that this agrees with the def of filteredness in 1-category theory.

Def Let κ be a regular cardinal and let $\mathcal{C}$ be an ∞ -category with κ -filtered colimits. We say that an object $X \in \mathcal{C}$ is κ -compact if

$$\mathrm{Map}_{\mathcal{C}}(X, -) : \mathcal{C} \rightarrow \mathbf{An}$$

preserves κ -filtered colimits. Write $\mathcal{C}^{\kappa} \subset \mathcal{C}$ for the full subcat spanned by the κ -compact objects.

We usually use the term compact instead of ω -compact.

Def Let κ be a regular cardinal and let $\mathcal{C}$ and $\mathcal{D}$ be ∞ -cats with κ -filtered colimits. A functor $F : \mathcal{C} \rightarrow \mathcal{D}$ is κ -finitary if F preserves κ -filtered colimits.

Def Let $\mathcal{C}_0$ be a small ∞ -category and let κ be a regular cardinal. The Ind_{κ} -completion of $\mathcal{C}_0$ is the universal ∞ -cat equipped with a functor $\gamma : \mathcal{C}_0 \rightarrow \mathrm{Ind}_{\kappa}(\mathcal{C}_0)$ satisfying the following properties:

(1) The ∞ -category $\mathrm{Ind}_{\kappa}(\mathcal{C}_0)$ has κ -filtered colimits.

(2) For any ∞ -category D with k -filtered colimits, precomp with y defines an equivalence

$$\mathrm{Fun}^{k\text{-fin}}(\mathrm{Ind}_k(\mathcal{C}_0), D) \xrightarrow{\sim} \mathrm{Fun}(\mathcal{C}_0, D).$$

Construction. $\mathrm{Ind}_k(\mathcal{C}_0) \subset \mathrm{Psh}(\mathcal{C}_0)$ is the smallest full subcategory of $\mathcal{C}$ containing the image of the Yoneda embedding and closed under k -filtered colimits, and y is the Yoneda embedding.

Def. Let $\mathcal{C}$ be an ∞ -category with k -filtered colimits. We say that $\mathcal{C}$ is k -compactly generated if the unique k -finitary extension of the inclusion $\mathrm{Ind}_k(\mathcal{C}^k) \rightarrow \mathcal{C}$ is an equivalence.

An ∞ -category $\mathcal{C}$ is accessible if $\mathcal{C}$ is k -compactly gen for some reg cardinal k . A functor between accessible ∞ -cats is accessible if it is k -finitary for some regular cardinal k .

Ex. A_n is compactly generated. For an E_∞ -ring A , the ∞ -cat Mod_A is compactly generated.

Ex. If X is a locally compact Hausdorff space, then $\text{Sh}(X, A_n)$ is N_1 -compactly generated. An observation of Neeman is that if X is a non-compact connected manifold, then the only compact object of $\text{Sh}(X, A_n)$ is the initial object. In particular, $\text{Sh}(X, A_n)$ is not compactly generated.

Def. An ∞ -category $\mathcal{C}$ is presentable if $\mathcal{C}$ is accessible and has all colimits.

Facts about presentable ∞ -categories

Thm. A presentable ∞ -category has all limits.

Adjoint Functor Thm. Let $F: \mathcal{C} \rightarrow \mathcal{D}$ be a functor between presentable ∞ -categories. Then:

- (1) F is a left adjoint if and only if F preserves colimits.
- (2) F is a right adjoint if and only if F is accessible and preserves limits.

Ntn. We write $\mathcal{P}\mathcal{r}^L \subset \mathcal{Cat}_\infty$ for the ∞ -category of presentable ∞ -categories and left adjoints. Similarly, $\mathcal{P}\mathcal{r}^R \subset \mathcal{Cat}_\infty$ denotes the subcategory of presentable ∞ -categories and right adjoints.

Note that

$$(\mathcal{P}\mathcal{r}^L)^{\mathrm{op}} \simeq \mathcal{P}\mathcal{r}^R$$

Thm. The ∞ -categories $\mathcal{P}\mathcal{r}^L$ and $\mathcal{P}\mathcal{r}^R$ preserve limits and the forgetful functors $\mathcal{P}\mathcal{r}^L \rightarrow \mathcal{Cat}_\infty$ and $\mathcal{P}\mathcal{r}^R \rightarrow \mathcal{Cat}_\infty$ preserve them.

Ex. If X is a stack that can be written as a small colimit of representables, then $\mathcal{Q}\text{Coh}(X)$ is presentable.

The tensor product

Def. Let $\mathcal{C}, \mathcal{D} \in \mathbf{Pr}^L$. The tensor product $\mathcal{C} \otimes \mathcal{D}$ is the universal presentable ∞ -category equipped with a functor $\otimes: \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \otimes \mathcal{D}$ satisfying

- (1) $\otimes: \mathcal{C} \times \mathcal{D} \rightarrow \mathcal{C} \otimes \mathcal{D}$ preserves colimits separately in each variable
- (2) If $\mathcal{E} \in \mathbf{Pr}^L$, then restriction along $\otimes$ defines an equivalence

$$\begin{array}{ccc} \mathrm{Fun}^{\mathrm{colim}}(\mathcal{C} \otimes \mathcal{D}, \mathcal{E}) & \xrightarrow{\sim} & \mathrm{Fun}^{\mathrm{colim}, \mathrm{colim}}(\mathcal{C} \times \mathcal{D}, \mathcal{E}) \\ \uparrow & & \uparrow \\ \text{colimit-preserving} & & \text{preserves colims sep. in} \\ & & \text{each variable} \end{array}$$

Fact. The tensor product refines to a sym mon structure on $\mathbf{Pr}^L$ with unit $\mathbf{An}$.

Construction. For $\mathcal{C}$ and $\mathcal{D}$ presentable, we have

$$\mathcal{C} \otimes \mathcal{D} \simeq \mathrm{Fun}^{\mathrm{lim}}(\mathcal{C}^{\mathrm{op}}, \mathcal{D}) \simeq \mathrm{Fun}^{\mathrm{lim}}(\mathcal{D}^{\mathrm{op}}, \mathcal{C})$$

This makes the functionality in right adjoints clear

$$\mathcal{C} \otimes (-) \simeq \text{Fun}^{\text{lim}}(\mathcal{C}^{\text{op}}, -) : \text{Pr} \mathcal{R} \rightarrow \text{Pr} \mathcal{R}$$

is just given by post-composition.

Ex. If $\mathcal{C} \hookrightarrow \mathcal{D} \rightarrow \mathcal{E}$ is a fully faithful right adjoint of presentable ∞ -categories, then $\mathcal{C} \otimes \mathcal{D} \rightarrow \mathcal{C} \otimes \mathcal{E}$ is fully faithful.

Ex. If A and B are $\mathbb{E}_{\infty}$ -rings, then

$$\text{Mod}_A \otimes \text{Mod}_B \simeq \text{Mod}_{A \otimes B}$$

Ex. If $\mathcal{C}_0$ is a small ∞ -category,

$$\text{Fun}(\mathcal{C}_0, \mathcal{A}) \otimes \mathcal{E} \simeq \text{Fun}(\mathcal{C}_0, \mathcal{E})$$

Observation. The ∞ -category $\text{Catg}(\text{Pr}^{\text{L}})$ is equivalent to the subcategory of $\text{CMon}(\text{Cat}_{\infty})$ with objects presentable ∞ -cats $\mathcal{C}$ with a sym. mon. structure $\otimes : \mathcal{C} \times \mathcal{C} \rightarrow \mathcal{C}$ that preserves colims sep. in each variable. Morphisms are symmetric

monoidal colimit-preserving functors

Objects of $\mathcal{C}Alg(\mathbf{Pr}^L)$ are called presentably symmetric monoidal ∞ -categories.

Ex. Both $(\mathbf{An}, \times)$ and $(\mathbf{Sp}, \otimes)$ are idempotent algebras in $\mathbf{Pr}^L$.

Sheaves and $*$ -functionality

Proposition Let X be a top space and $\mathcal{E}$ a presentable ∞ -category. Then the inclusion

$$\mathrm{Sh}(X, \mathcal{E}) \hookrightarrow \mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathcal{E})$$

admits a left adjoint L called **Sheafification**. Moreover

(1) $\mathrm{Sh}(X, \mathcal{E})$ is presentable

(2) If $\mathcal{E}$ is compactly generated, L is left exact

(3) There is a natural equivalence $\mathrm{Sh}(X, \mathcal{A}_n) \otimes \mathcal{E} \xrightarrow{\sim} \mathrm{Sh}(X, \mathcal{E})$
compatible with $\mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathcal{A}_n) \otimes \mathcal{E} \xrightarrow{\sim} \mathrm{Fun}(\mathrm{Open}(X)^{\mathrm{op}}, \mathcal{E})$

Notation Let $f: X \rightarrow Y$ be a map of top spaces. We write

$$f_*: \mathrm{PSh}(X, \mathcal{A}_n) \longrightarrow \mathrm{PSh}(Y, \mathcal{A}_n)$$

for the functor $f_*(F)(V) := F(f^{-1}(V))$.

- > Note that f_* carries sheaves to sheaves
- > Note that f_* preserves limits and k -filtered colimits for any k such that the inclusions $\text{Sh}(X, \mathcal{A}_n) \subset \text{PSh}(X, \mathcal{A}_n)$ and $\text{Sh}(Y, \mathcal{A}_n) \subset \text{PSh}(Y, \mathcal{A}_n)$ do
- > So f_* admits a left adjoint f^* . Moreover, for $V \subset Y$ open,

$$f^*(\mathbb{Z}(V)) \simeq \mathbb{Z}(f^{-1}(V))$$

Consequence For $\mathcal{E}$ any presentable ∞ -category, by tensoring with $\mathcal{E}$ we obtain an adjunction

$$f^* : \text{Sh}(Y, \mathcal{E}) \rightleftarrows \text{Sh}(X, \mathcal{E}) : f_*$$

Moreover, f_* is given by the same formula as before

Observation. Let $j : U \hookrightarrow X$ be an open immersion. Then the functor $\text{Sh}(X, \mathcal{E}) \rightarrow \text{Sh}(U, \mathcal{E})$ given by

$$F \mapsto [V \subset U \mapsto F(V)]$$

is left adjoint to j_* . Moreover, this functor preserves limits.
Hence j^* admits an additional left adjoint $j_! : \mathrm{Sh}(U, \mathcal{E}) \rightarrow \mathrm{Sh}(X, \mathcal{E})$.

Observe. By the above, $j^* j_* \simeq \mathrm{id}$. Hence j_* is fully faithful.
As a consequence, so is $j_!$.